

Data Mining & Machine Learning
M.Sc. in Bionics Engineering
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Project Report

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**Project Title: Locomotion mode Recognition From
Exoskeleton-Embeddable Sensors**

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Abstract

This project addresses locomotion mode recognition for the control loop of a lower-limb exoskeleton in a rehabilitation setting. Six locomotion modes are considered: idle, walking, stair ascent/descent, and ramp ascent/descent, using only sensors that can realistically be embedded in the device: four inertial measurement units and three goniometers on the right side of the body. Data come from the Camargo dataset, segmented into 250 ms windows and summarized by 555 statistical features. Three classifier families are compared under subject-independent (leave-one-subject-out) cross-validation — a Decision Tree, a Random Forest, and XGBoost — with mutual-information-based feature selection fitted on each fold's training data and hyperparameters selected by mean LOSO macro-F1, and their differences tested for statistical significance with a paired Wilcoxon signed-rank test. XGBoost achieves the highest macro-F1 (0.8389 in cross-validation, 0.8535 on four held-out subjects) and is the delivered model; both ensembles significantly outperform the single tree, and XGBoost's advantage over Random Forest, while smaller, is also significant. SHAP-based explainability shows that sagittal-plane joint-angle features rank among the most important (22.3% of total SHAP importance), consistent with the biomechanics of the task. Error analysis identifies two distinct failure modes: shallow ramps are geometrically similar to level walking and are frequently confused with it, while idle's F1 drops sharply on transition-dominated windows, consistent with residual movement persisting into the window. The idle-to-walk confusion is judged the most safety-critical, since it could trigger unwanted assistance while the user is stationary.

1. Introduction

The application domain is wearable exoskeletons for rehabilitation, specifically a Lower-Limb Exoskeleton (LLE) with IMUs on the trunk, thigh, shank, and foot, and goniometers at the hip, knee, and ankle. We focus on rehabilitation rather than the commercial/industrial setting because it constrains the environment and admits a well-defined activity set, while commercial/industrial exoskeletons require broader generalization. The underlying task is locomotion mode recognition (LMR), which can operate as part of the high-level control layer of a lower-limb exoskeleton: the recognized locomotion mode determines the device's operating mode and can condition the behavior of the downstream control layers [1]. Locomotion transitions are particularly challenging, since the controller must adapt its behavior as the user moves from one locomotion mode to another. Recognition errors or delays during these transitions may therefore lead to inappropriate changes in the assistance provided by the exoskeleton [2].

This project aims to:

1. Build a 6-class locomotion-mode recognizer using only sensors embeddable in a real exoskeleton (29 IMU and goniometer channels, right side only), validated in a subject-independent setting.
2. Compare a single-tree classifier (Decision Tree) against a bagging ensemble (Random Forest) and a boosting one (XGBoost), and test whether the difference in performance is statistically significant.
3. Explain the winning model at both the global and local level using SHAP, and analyze its errors to identify where misclassification is most dangerous for exoskeleton control.

2. Problem

The problem is a six-class classification task: idle (no locomotion), walking (3 self-selected speeds), stair ascent/descent (4 step heights), and ramp ascent/descent (6 inclinations). The signal is segmented into 250 ms windows with 50% overlap (125 ms stride), from 29 channels coming from 7 sensors (4 IMUs + 3 goniometers) placed on the right side of the body.

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	Multiclass classification
Dataset	Camargo et al. (2021), EPIC Lab — 22 subject, 2.990 trial, 9.917.736 samples (13,8 h) [3]
Target variable	Locomotion Mode
Features	555: 15 statistics × 37 channels (29 raw + 8 derived magnitude channels)
Missing values	0.00143% of samples overall; per-class breakdown below
Class distribution	idle 35.32% · walk 28.30% · ramp_descent 12.85% · ramp_ascent 12.45% · stair_ascent 5.67% · stair_descent 5.41% (6.5× imbalance)
Evaluation metric(s)	F1-score (macro)
Train / Val / Test	18 development subjects (18-fold LOSO) / 4 held-out subjects (test)

2.1 Exploratory Data Analysis

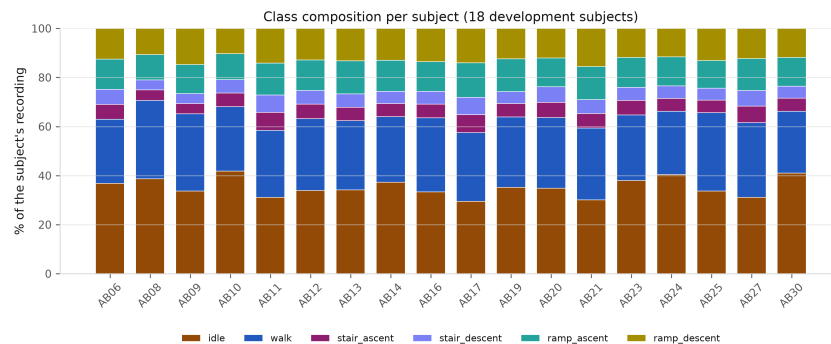


Figure 1. Class Composition per subject

Figure 1 shows that each subject (i.e., each LOSO fold) has a fairly different class composition.

Table 2 shows that the percentage of missing values is negligible for every class.

Figure 2 shows the Pearson correlation matrix between the 29 raw channels, grouped by sensor location. The strongest correlations sit inside the same block, between channels on the same segment or joint; correlation across different segments is lower.

Class	Missing
stair_descent	0.00703%
idle	0.00240%
walk	0.00069%
ramp_ascent	0.00001%
stair_ascent, ramp_descent	0%
Overall	0.00143%

Table 2. Missing-value rate per class

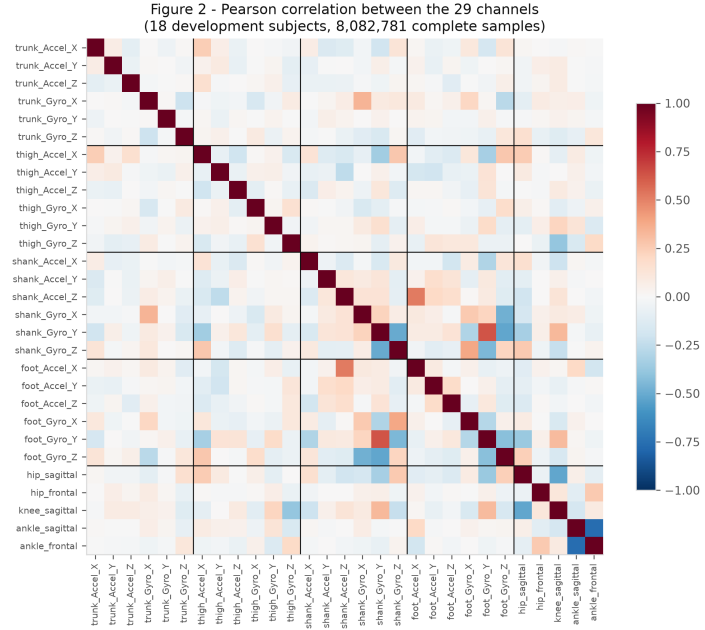


Figure 2. Channel Correlation

2.2 Feature Description

Each 250 ms window (50 samples at 200 Hz) is summarized by **15 statistics** computed independently on **37 channels**, for a total of **555 features**. Every feature is named `{channel}_{statistic}`.

Table 3. The 37 channels behind the 555 features.

Group	Channels
IMU acc. & gyro. (24)	<code>{trunk,thigh,shank,foot}_{Accel,Gyro}_{X,Y,Z}</code>
Goniometer (5)	<code>{hip,knee,ankle}_sagittal, {hip,ankle}_frontal</code>
Derived — vector magnitude (8)	<code>{trunk,thigh,shank,foot}_{Accel,Gyro}Mag = $\sqrt{X^2 + Y^2 + Z^2}$ of the corresponding accel./gyro. axes</code>

Table 4. The 15 per-channel, per-window statistics.

Statistics	Definition
mean, std	Sample mean and standard deviation of X
min, max, range	Minimum, maximum, and $\max(X) - \min(X)$
rms, median, iqr	Root mean square $\sqrt{\text{mean}(X^2)}$; median; interquartile range $Q_{75} - Q_{25}$
skew, kurt	Skewness $\text{mean}(z^3)$ and excess kurtosis $\text{mean}(z^4) - 3$, with $z = (X - \text{mean})/\text{std}$
mad_diff, denergy	Mean absolute and mean squared first difference, $\text{mean}(\Delta X)$ and $\text{mean}(\Delta X^2)$ — short-term dynamics descriptors
blo, bmid, bhi	Share of spectral energy in (0,10] Hz, (10,25] Hz, (25,100] Hz, from $ \text{FFT}(X) ^2$ normalized by total positive-frequency energy

The feature set combines time-domain statistical descriptors and frequency-domain energy descriptors. Features such as minimum, maximum, mean, standard deviation, and RMS have been used in locomotion-mode recognition from kinematic and inertial signals in active orthoses

and exoskeletons [2]. More broadly, statistical descriptors such as range, median, interquartile range, skewness and kurtosis, together with spectral-energy features, are commonly used in wearable-sensor HAR [4].

3. Pipeline and Experiments

3.1 Data transposition

The dataset is stored in `.mat` files, not in the classical `struct` format but as `table/timetable` objects, so we first had to convert them into `pandas.DataFrame` using the `mat-io` library.

3.2 Preprocessing

1. **Data retained:** Of the sensor groups recorded by the dataset, only three are used: the IMUs, the goniometers, and the conditions folder (trial labels and metadata). The remaining groups — EMG, force plates, marker-based motion capture, and inverse kinematics/dynamics — are dropped because none of them could realistically be embedded in an exoskeleton.
2. **Data alignment:** The goniometers are sampled at 1000 Hz against 200 Hz for the IMUs, which therefore act as the master clock. Each IMU timestamp is matched to its nearest goniometer sample, within a tolerance of ± 2.5 ms — a nearest-neighbor snap, not an average over the block of 5 goniometer samples in that interval, so it introduces no smoothing. IMU timestamps with no goniometer sample within tolerance are discarded.
3. **Label Mapping:** Raw labels include five steady states and ten transitions between them (e.g. `walk-stairascent`, `stairascent-walk`). Each transition sample takes the label of its destination state, not a separate transition class.
4. **Windowing:** Each trial is segmented into 250 ms windows (50 samples at 200 Hz), consistent with window durations reported for mechanical-sensor-based locomotion mode recognition in active orthoses and exoskeletons [2]. Consecutive windows use a 125 ms stride (50% overlap), chosen as a compromise between prediction frequency and computational cost. A window's label is the mode of its 50 per-sample labels (already mapped onto the 6 target classes); ties are broken by the last sample, on the assumption that the most recent state is the more relevant one.
5. **Invalid samples:** A window is discarded if any of its 50 samples is invalid. Discarding introduces no meaningful bias, since the number of affected windows (118) is negligible compared to the total.
6. **Subject split:** All prior steps use the full dataset. From here on, every decision that could bias model selection is confined to the 18 development subjects; the 4 held-out subjects are chosen so that none share a recording session (and its calibration conditions) and each falls in a different quartile of per-subject data volume. Feature selection and hyperparameter tuning use 18-fold LOSO cross-validation on the development subjects; the four held-out subjects are reserved for the final test of the three tuned models.

3.3 Feature Selection and Engineering

Feature selection ranks the 555 candidate features by Mutual Information, recomputed inside each of the 18 LOSO folds using only that fold's training rows to avoid leakage.

Each model's grid includes a shared feature-count axis $k \in \{75, 150, 350\}$, so k is treated as a hyperparameter; $k_{max} = 555$ (all features) was discarded to keep computational cost down.

Figure 3 shows the top-ranked features by mean Mutual Information across folds: 18 of the top

20 are IMU-derived, against 2 from goniometers, though the picture is more balanced when aggregated by sensor group.

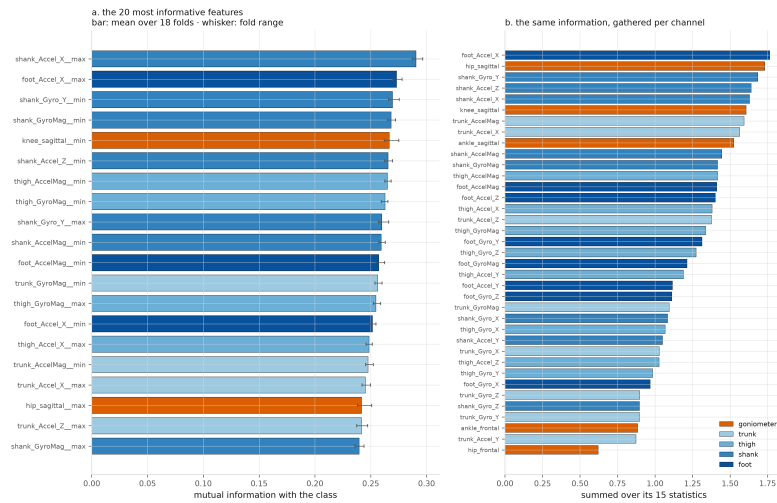


Figure 3. Feature ranked based on Mutual information

3.4 Model Selection and Cross-Validation

Three model families are compared: a single Decision Tree, a Random Forest (bagging), and XGBoost (boosting), the latter two assessed against the tree baseline. All configurations are evaluated with Leave-One-Subject-Out (LOSO) cross-validation on the 18 development subjects (17 for training, 1 for validation per fold); the four held-out subjects are excluded from fitting and hyperparameter selection. Within each fold, the `idle` subsample and Mutual Information ranking are computed on the training data only and reused across candidate configurations; the validation subject keeps all its windows. All the three classifier use Z-normalized features (fitted on the training fold). Training `idle` windows are downsampled to 40% of their count (mainly for compute cost); remaining imbalance is handled via `class_weight="balanced"` (Decision Tree), `class_weight="balanced_subsample"` (Random Forest), and sample weights from `compute_sample_weight("balanced", y_train)` passed through `sample_weight` (XGBoost). Three model families are compared under an identical protocol: Decision Tree, Random Forest, and XGBoost, each grid-searched (Table 5) under 18-fold LOSO cross-validation, selected and reported on macro-F1.

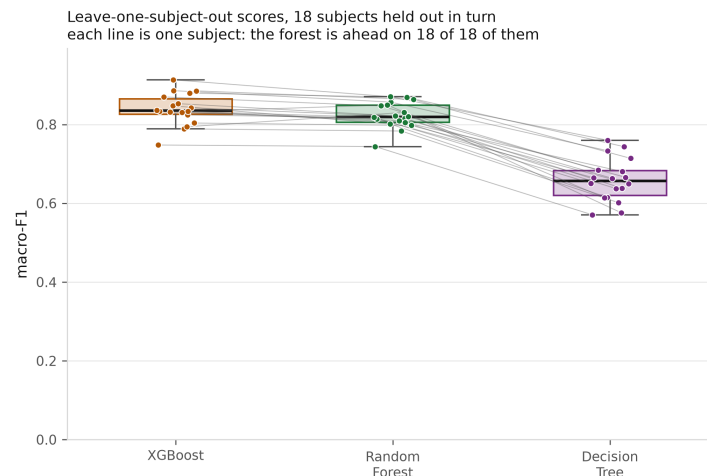


Figure 4. Cross-Validated scores of the models

Figure 4 shows per-subject variability: AB11 has the lowest macro-F1 under all three configurations, so subsequent comparisons pair scores by validation subject.

3.5 Hyperparameter Optimization

Each family is tuned via grid search over the shared axis $k \in \{75, 150, 350\}$ and the family-specific axes in Table 5 (non-nested, same 18 LOSO folds as above), selecting the configuration with the highest mean macro-F1. For XGBoost, each $(k, \text{learning rate})$ pair is fit once per fold for 600 rounds, and the five candidate round counts are read from that single fit via `iteration_range`.

Table 5. Hyperparameter search spaces. k is shared with feature selection (§3.3)

Model	Grid
Decision Tree (27 configs)	<code>max_depth</code> $\in \{15, 20, 30\}$, <code>min_samples_leaf</code> $\in \{1, 10, 50\}$, $k \in \{75, 150, 350\}$
Random Forest (18 configs)	<code>n_estimators</code> $\in \{100, 200\}$, <code>min_samples_leaf</code> $\in \{1, 5, 30\}$, $k \in \{75, 150, 350\}$; <code>max_features</code> ="sqrt" and <code>max_depth</code> =None fixed
XGBoost (30 configs)	<code>learning_rate</code> $\in \{0.1, 0.3\}$, $k \in \{75, 150, 350\}$, <code>n_estimators</code> read at $\{100, 200, 300, 400, 600\}$ from a single fit via <code>iteration_range</code> ; <code>max_leaves</code> =63.

The winning configurations (mean $\pm$ SD macro-F1 across the 18 folds) are:

Decision Tree, `max_depth`=15, `min_samples_leaf`=1, $k=350$ (0.6590 ± 0.0543);

Random Forest, `n_estimators`=200, `min_samples_leaf`=5, $k=350$ (0.8232 ± 0.0326);

XGBoost, `learning_rate`=0.3, $k=350$, `n_estimators`=600 (0.8389 ± 0.0400).

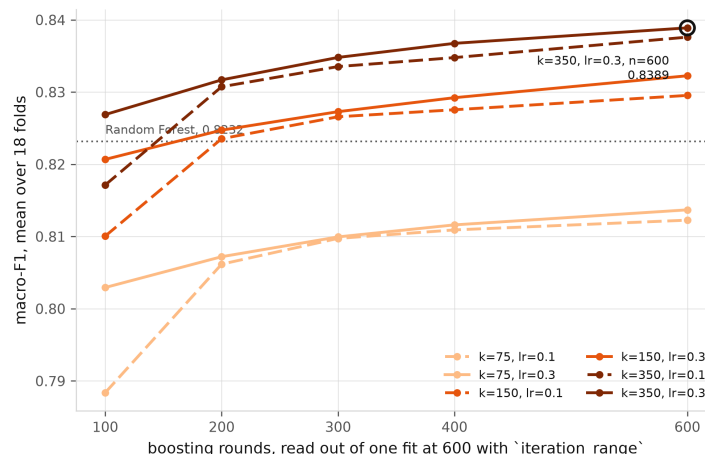


Figure 5. Validation curve XGBoost

3.6 Statistical Tests

Table 6. Exploratory paired comparisons of the selected configurations across 18 LOSO folds.

Comparison	Δ	W	Holm p	Wins
XGBoost – Random Forest	+0.0157	24	0.00560	16/18
XGBoost – Decision Tree	+0.1799	0	0.0000229	18/18
Random Forest – Decision Tree	+0.1642	0	0.0000229	18/18

The selected configurations are compared using two-sided paired Wilcoxon signed-rank tests on the 18 LOSO macro-F1 scores, paired by validation subject, with Holm correction across the three comparisons at $\alpha = 0.05$ (Table 6). W is the smaller of the positive/negative rank sums. All three comparisons remain significant after Holm correction (XGBoost over Random Forest in 16/18 folds, both ensembles over the single tree in 18/18): this is exploratory evidence, since configurations were selected on the same CV scores and the LOSO folds' training sets overlap substantially — Holm corrects only for the three comparisons, not for these two effects.

3.7 Final Model Training

Each selected configuration is retrained on all 18 development subjects and evaluated on the four held-out subjects; XGBoost, with the highest mean LOSO macro-F1, is the final model. The `idle` subsample, Mutual Information ranking, and feature scaler are all refit on the pooled development data only (training set: 319,651 $\rightarrow$ 253,235 windows after subsampling); no held-out observation is used in fitting. Held-out macro-F1 is 0.8535 (XGBoost), 0.8426 (Random Forest), and 0.6807 (Decision Tree) — respectively +0.0146, +0.0194, +0.0217 above the corresponding mean LOSO scores. These differences are descriptive rather than a leakage check: CV averages subject-specific scores whereas the test score pools windows across the four held-out subjects, which also differ in size and identity from any single fold. Subject separation during fitting and selection is what protects against leakage, not the sign of this gap; with only four held-out subjects, conclusions about generalization remain limited.

4. Results and Discussion

4.1 Performance Evaluation

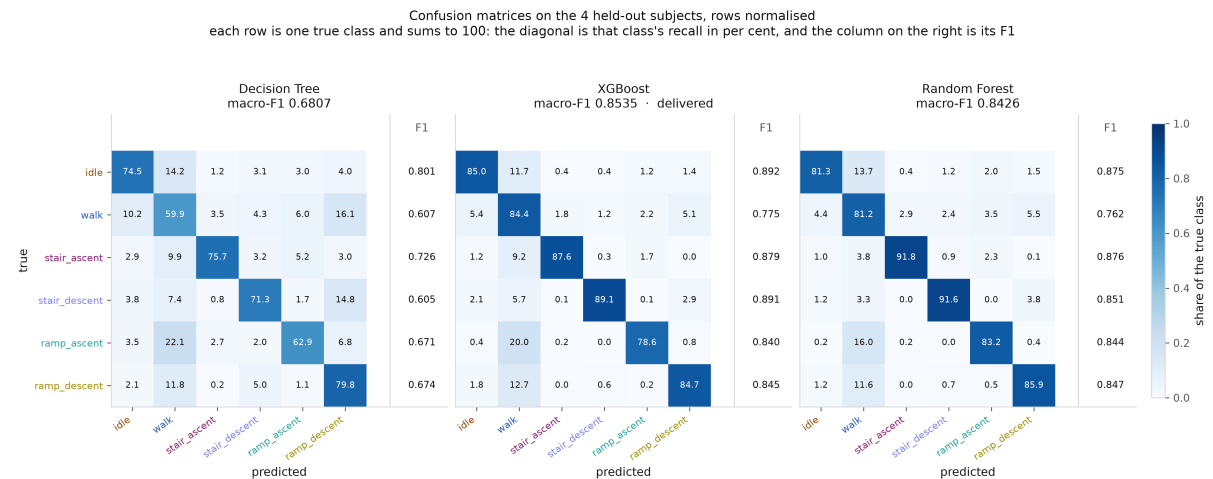


Figure 6. Confusion matrix of the held out test

The confusion matrix shows that `walk` acts as a catch-all class rather than a poorly recognized activity: the largest errors are `ramp_ascent`, `ramp_descent`, and `idle` classified as `walk`, the last being the most safety-critical since it may trigger assistance while the user is stationary. For `idle`, performance is substantially lower on transition-dominated windows than on the remaining windows (OOF F1: 0.345 versus 0.913; Table 7). Residual movement during transitions to standing, mapped to `idle`, is one plausible explanation. However, most misclassified `idle` windows belong to the non-transition-dominated group.

Table 7. Per-class F1 on stationary vs. transition windows out of fold

Class	F1 (stat.)	F1 (trans.)	Loss
idle	0.913	0.345	+0.568
ramp_descent	0.809	0.722	+0.087
stair_ascent	0.894	0.841	+0.053
ramp_ascent	0.884	0.847	+0.037
stair_descent	0.872	0.838	+0.034
walk	0.774	0.771	+0.003

Figure 7 confirms the model ranking across decision thresholds: XGBoost achieves the highest macro average precision (0.922), followed by Random Forest (0.903) and Decision Tree (0.590). Its advantage over Random Forest concentrates mainly on the stair classes, especially stair descent; despite their low prevalence, both stair classes reach an AP of 0.944, showing that rarity alone does not determine performance. Conversely, *walk* has the lowest XGBoost AP (0.838), consistent with its catch-all role in the confusion matrix.

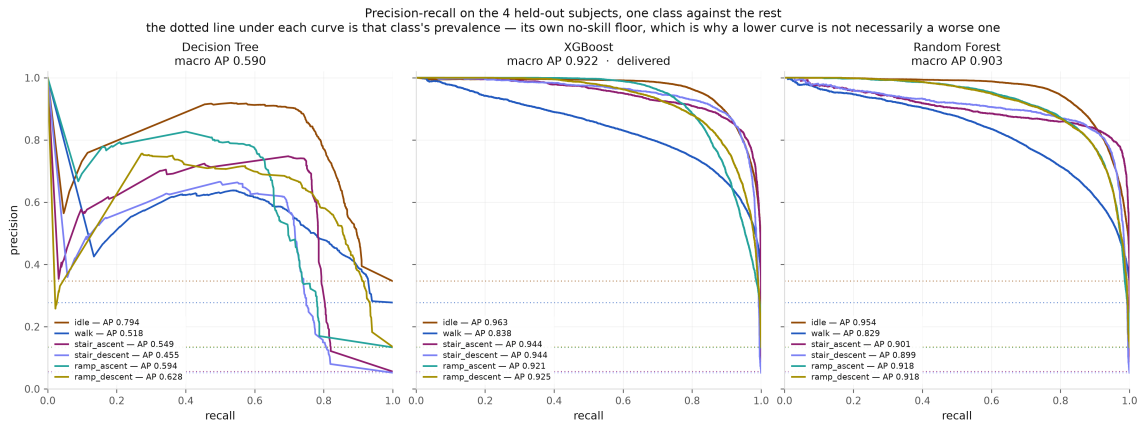


Figure 7. Precision - Recall graph

4.2 Model Explainability

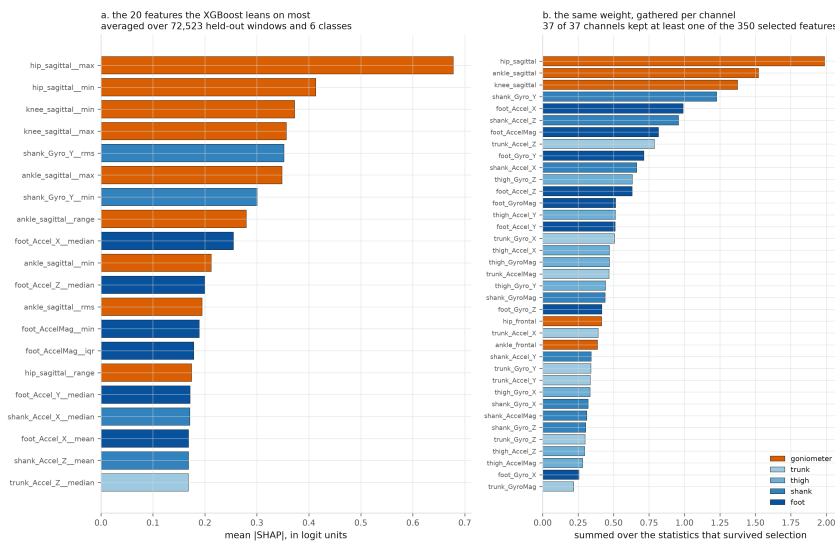


Figure 8. mean global |SHAP|

Figure 8 shows the global mean absolute SHAP values of the delivered XGBoost model; its top 20 features differ substantially from the Mutual Information ranking, consistent with MI being used only for feature selection, not as the splitting criterion within the trees. From a biomechanical perspective, features derived from hip, knee, and ankle angles in the sagittal plane occupy the first three positions once aggregated by channel, together accounting for 22.3% of total SHAP importance (26% across all five goniometer channels, including the two frontal-plane ones); the remaining importance is spread over the 32 IMU channels, none individually dominant consistent with, but not proof of, a biomechanical explanation for the task.

Table 8. Absolute and invariant statistics: share of the selected features against share of the SHAP.

Family	Features	Expected	Final
Absolute	164	46.9%	70.6%
Invariant	186	53.1%	29.4%

Features can also be grouped by how their statistics react to a constant offset of the signal: *absolute* ones (mean, min, max, median etc.) shift with it, *invariant* ones (standard deviation, range, IQR, etc.) do not. Absolute statistics take 70.6% of the SHAP weight against the 46.9% their number would give them; the invariant family mirrors it, 186 of the 350 selected features for under a third of the weight. On a goniometer such a statistic is a joint angle, and maximum hip flexion separating a stair from standing still is physiology; on an IMU channel, instead, it depends largely on how the sensor was mounted on that subject.

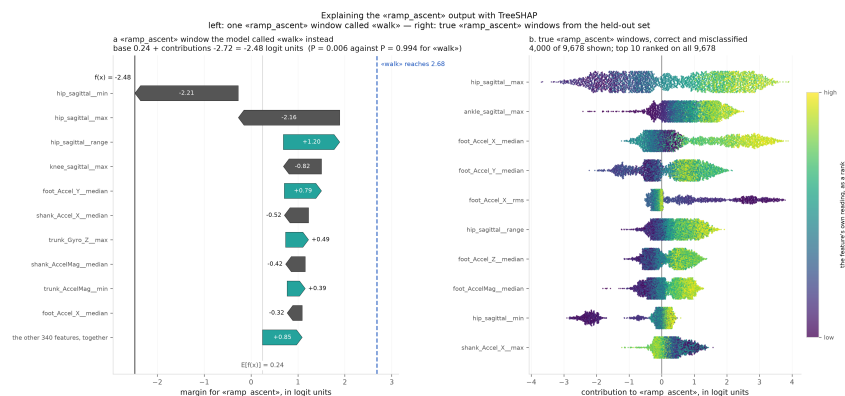


Figure 9. Misclassified ramp_ascent window and a beeswarm restricted to true ramp_ascent windows.

Figure 9 combines a local and a global explanation of the same failure mode. Panel (a) is a waterfall plot for one held-out ramp_ascent window that the model misclassified as walk: starting from the base rate, it shows how each feature pushes the ramp_ascent logit up or down, down to the model's final (wrong) call. Panel (b) generalizes this view with a beeswarm over all 9,678 true ramp_ascent windows, correct and incorrect alike: each dot is one window's SHAP contribution for a given feature, colored by that feature's own value, so the two panels can be read together — one concrete failure case against the pattern it belongs to.

4.3 Error Analysis

Table 9 shows that incline angle strongly affects the share of windows misclassified as walk, most markedly for ramp_ascent. Figure 9 helps interpret this: in the selected ramp_ascent window misclassified as walk, hip_sagittal_min and hip_sagittal_max provide the largest negative contributions to the ramp_ascent score, and the beeswarm — restricted

to true `ramp_ascent` windows, correct and incorrect alike — shows that lower values of `hip_sagittal__max` and `ankle_sagittal__max` tend to contribute negatively to the same score. Together with the greater ramp-to-walk confusion at low inclinations (Table 9), this suggests that less pronounced joint flexion may make shallow ramps harder for the model to distinguish from level walking.

Table 9. Ramp window classified as level walking

Incline	Ascent n	Ascent $\rightarrow$ walk	Descent n	Descent $\rightarrow$ walk
5.200°	1686	38.3%	1679	22.3%
7.795°	1512	31.5%	1524	11.4%
9.207°	1561	23.6%	1564	10.4%
10.989°	1590	12.7%	1622	9.6%
12.416°	1593	9.9%	1686	8.2%
18.000°	1736	4.8%	1712	13.9%

5. Conclusions

XGBoost achieves the highest macro-F1 both in LOSO cross-validation and on the held-out test subjects. Sagittal-plane goniometer statistics rank among the most important features (22.3% of total SHAP importance), consistent with the classifier relying on biomechanical signal.

The main errors involve `walk`, confused with `ramp_ascent` and `idle`. The `idle` $\rightarrow$ `walk` confusion was judged the most safety-critical, since it could trigger unwanted assistance while the user is stationary; `ramp_ascent` $\rightarrow$ `walk` at low inclination is also worth improving but plausibly lower-risk if the controller issues similar commands for both — this is an application-motivated judgment, not a measured severity ranking.

(1) 6-class recognizer: Held-out macro-F1 (0.8535) is close to the mean LOSO estimate (0.8389), despite the two using different aggregations (CV averages per-subject scores; the held-out score pools windows across four subjects) — consistent with transfer to unseen subjects, though leakage is controlled by design (subject separation during fitting and selection), not established by this gap; with only four held-out subjects, generalization conclusions remain limited.

(2) Model comparison: Paired Wilcoxon tests with Holm correction show Decision Tree < Random Forest < XGBoost, all significant at $\alpha = 0.05$. These are exploratory comparisons among configurations selected on the same LOSO folds, favoring this ranking under the adopted procedure.

(3) Explainability: SHAP helps interpret why the classifier struggles most on `walk`, `idle`, and `ramp_ascent`: a biomechanical explanation for the ramp classes and residual movement carried into `idle`’s transition windows are both plausible, not demonstrated, causes.

Limitations and future work. With only four held-out subjects, the result cannot rule out residual overfitting. The scenario is also constrained by design: rehabilitation only, six locomotion modes, right-side sensors only motivating tests on a broader population. The 250 ms/125 ms window was chosen for compute cost rather than discriminability, so other lengths may be worth studying; and `idle`’s lower performance on transition windows motivates finer-grained transition labels or soft/uncertain outputs as a further direction.

References

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